



Cardiff
Metropolitan
University

Prifysgol
Metropolitan
Caerdydd

School of Technologies

Assessment Brief

Module Code

Module Title

CIS6008

Analytics and Business Intelligence

Academic Year

Semester

2026

2

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Assessment Details

Assessment title	Abr.	Weighting
Evaluating the Effectiveness and Efficiency of Statistical and Geospatial Tools in Generating Business Intelligence for Informed Decision-Making in the Development of the Civil Aviation Sector in Sri Lanka.	PRE1	20%

Pass marks are 40% for undergraduate work and 50% for postgraduate work unless stated otherwise.

Task/assessment brief:

Purpose:

This business proposal presentation aims to assess students' ability to design and effectively communicate the application of Statistical and Geographic Information Systems (GIS) tools, techniques, and methodologies to generate actionable intelligence for informed decision-making in both public and private sector institutions in Sri Lanka and globally.

The task requires students to demonstrate how these analytical approaches can be applied to identify solutions to complex problems, uncover emerging opportunities, and support sustainable development across various sectors. Furthermore, the proposal should highlight how senior administrative and policy-level decision-makers can leverage statistical analysis and geospatial technologies to enhance efficiency, optimize resource utilization, and maximize organizational and economic benefits.

The application of these tools is expected to facilitate evidence-based decision-making by reducing ambiguity, minimizing uncertainty, and promoting fairness and transparency in business and policy environments.

Tasks Introduction

Understand the given tasks based on Sri Lanka using data science domain associate with geospatial and non-geospatial data models. Apply relevant tools, techniques and methodologies found in business analytics subject relevant to the module scope and develop a business proposal presentation relevant to the topic of the task.

Task 1 – Business Proposal Presentation (100 Marks)

Develop a proposal presentation that critically evaluates how the effective application of statistical and geospatial tools, techniques, and methodologies can be utilized to generate Business Intelligence (BI) for informed decision-making in the improvement of Sri Lanka's Civil Aviation Sector.

The proposal should further emphasize how the strategic use of Business Intelligence can support evidence-based decision-making by the Ministry of Transport, Highways, Ports and Civil Aviation. In

particular, it should demonstrate how data-driven insights derived from statistical analysis and geospatial technologies can enhance sector performance, optimize resource allocation, improve operational efficiency, and contribute to increasing foreign exchange earnings for Sri Lanka.

Your presentation should fulfill following requirements;

- The introduction to the topic and the relevance to the module and application of business analytics to support the topic discussing matters. (10 Marks)
- Include research literature review relevant to the discussion. How clearly student has recognized various statistical and geospatial software applications available as open source or licensed in the market to support the discussion. The knowledge about latest statistical and GIS software applications (15 Marks)
- Discuss the importance of the topic as per global/local trends. Identification of the issues/opportunities and solutions related to the selected topic with the support of statistical, geospatial tools, technologies and methodologies. Feasibility of the proposal should also be emphasized. (25 Marks)
- Prepare clear and precise presentation slides to aid the discussion of the topic. Considered a logical story that the student leading the audience through. (20 Marks)
- Verbally explain and present the subject matter discussed in the presentation. Student's knowledge fluency of topic and business analytics subject domain is highly considered. (20 Marks)
- The sources of information included using Harvard referencing system (10 Marks)

(10+15+25+20+20=100 Marks) (LO 1)

Word count (or equivalent):

1000

This is a reflection of the effort required for the assessment. Word counts will normally include source code, any text, tables, calculations, figures, subtitles and citations. Reference lists and contents of appendices are excluded from the word count. Contents of appendices are not usually considered when determining your final assessment grade.

Academic or technical terms explained:

ABI – Analytics and Business Intelligence

BI - Business Intelligence

GIS – Geographical Information Systems

GPS - Global Positioning System

Develop - Put statistical and geo spatial gained knowledge in relevant to the scenario together with emphasis on creating a new proposal for solving problem or improving present status of a certain geography or an organization.

Submission Details

Submission Deadline:

This will be provided on the Moodle submission point.

Estimated Feedback Return Date

This will normally be 20 working days after initial submission.

Submission Time:

By 2.00pm on the deadline day.

Moodle/Turnitin:

Any assessments submitted after the deadline will not be marked and will be recorded as a non-attempt unless you have had an extension request agreed or have approved mitigating circumstances. See the School Moodle pages for more information on extensions and mitigating circumstances.

File Format:

The assessment must be submitted as a pdf document (save the document as a pdf in your software) and submit through the Turnitin submission point in Moodle.

Your assessment should be titled with your:

**student ID number, module code and assessment ID,
e.g. st12345678 CSE5013 WRIT1**

Feedback

Feedback for the assessment will be provided electronically via Moodle. Feedback will be provided with comments on your strengths and the areas which you can improve. View the [guidance](#) on how to access your feedback.

All marks are provisional and are subject to [quality assurance processes](#) and confirmation at the programme Examination Board.

Assessment Criteria

Learning outcomes assessed

- Demonstrate understanding of the leading technologies relating to business intelligence, data analysis, predictive and other analytical technologies (e.g. geospatial, social), and be able to apply them appropriately in real world scenarios.

Learning Outcomes covered from the presentation.

a) LO 1

Other skills/attributes developed

This includes elements of the Cardiff Met EDGE (Ethical, Digital, Global and Entrepreneurial skills) and other attributes developed in students through the completion of the module and assessment. These will also be highlighted in the module guidance, which should be read by all students completing the module. Assessments are not just a way of auditing student knowledge. They are a process which provides additional learning and development through the preparation for and completion of the assessment.

ETHICAL	Understanding the importance of adhering to the formal ethical practices when different sources data is extracted ,processed and disseminate.This is achieved via university ethical guidelines and procedures.
DIGITAL	Making use of vector and raster data models for geo spatial analysis and processing and signifying stitstically.
GLOBAL	Usage of extracted data related to varouse parts or scenarios of the world for statistical, geo spatial analysis and processing.Localizing findings as per requiremnents of organizations and individuals.
ENTREPRENEURIAL	Involvement of business analytics and business intelligence; motivate stiudents to inintate their own organizations for providing various solutions recommedations for corporate and individual customers.

Presentation needs to be done covering following areas,

- A brief introduction to the topic
- Importance of the topic as per global/local trends
- Various issues, opportunities related to the topic
- How those issues possible to overcome and opportunities to make use with the knowledge of business intelligence and analytics.
- Research literature review related to the discussion.
- What geospatial and non-geospatial data are used
- What statistical and geospatial tools, techniques and methodologies can be used.
- Conclusion & Recommendation(s)
- Referencing, citations and captions.

Guidelines for Presentation

- Minimum 10-15 slides
- Suitable slide theme

- Clearly, easily readable text and suitable graphics (Graphs, Charts etc.)
 - Limit the amount of words you have on each slide
 - Font
 - Suitable clear and readable font style
 - Not to use less than 20pt font
 - Titles -30-40pt
 - Body text- 20-30pt
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- The student need to name the presentation softcopy with his/her name with initials and the university number (not ICBT number), not your topic or some weird abbreviation of your topic that only you understand.
 - You are required to submit the soft copy of the presentation slides to university.
 - The student will not be judged on “glitz and glam” on his/her presentation slides. Therefore, student should not use too much animation or anything that will distract the audience from the main points of the presentation. Pictures, graphs /Charts are nice in order to demonstrate a point or show the audience what the student is talking about but should not put in extraneous pictures just for the sake of having pretty slides.
 - Strict time limits will need to be enforced on student’s presentations. Therefore, student need to have a proper practice before the final.
 - Referencing and in-text citation should be done strictly using Harvard Referencing System.

Marking/Assessment Criteria

Task	Poor < 40	Satisfactory 40 - 49	Good 50-59	Very Good 60 -69	Excellent 70 -100
Introduction to the topic (10)	The introduction to the topic and the relevance to the module is not satisfactory. The student has not understood the effective application of business analytics to support the topic discussing matters not satisfactory way.	The introduction to the topic and the relevance to the module is satisfactory. The student has understood the effective application of business analytics to support the topic discussing matters better way.	The introduction to the topic and the relevance to the module is good. The student has understood the effective application of business analytics to support the topic discussing matters good way.	The introduction to the topic and the relevance to the module is very good. The student has understood the effective application of business analytics to support the topic discussing matters very good way.	The introduction to the topic and the relevance to the module is excellent. The student has understood the effective application of business analytics to support the topic discussing matters excellent way.
The sources of information (10)	The sources of information included poor way. The Citations and Referencing was not done as per Harvard referencing system satisfactory way covering all major domains of the presentation.	The sources of information better way included. The Citations and Referencing done as per Harvard referencing system satisfactory way covering all major domains of the presentation.	The sources of information good way included. The Citations and Referencing done as per Harvard referencing system better way covering all major domains of the presentation.	The sources of information very good way included. The Citations and Referencing done as per Harvard referencing system better way covering all major domains of the presentation.	The sources of information very well included. The Citations and Referencing done as per Harvard referencing system excellent way covering all major domains of the presentation.
Presenation slides (20)	poor and not satisfactory level presentation slides are developed not satisfactory level aid the discussion of the topic.Poor logical story that the student leading the audience through. Slides contain some required volume of information. Everything on the slide relevant to what student is explaining.	Clear and satisfactory level presentation slides are developed satisfactory level aid the discussion of the topic.satisfactory level developed logical story that the student leading the audience through. Slides contain required volume of information. Everything on the slide relevant to what student is explaining.	Clear and good presentation slides are developed to aid the discussion of the topic. Developed good logical story that the student leading the audience through. Slides contain required volume of information. Everything on the slide relevant to what student is explaining.	Clear and very good precise presentation slides are developed better manner to aid the discussion of the topic.better way developed logical story that the student leading the audience through. Slides contain required volume of information. Everything on the slide relevant to what student is explaining.	Clear and excellent precise presentation slides are developed excellent manner to aid the discussion of the topic.Very well developrd logical story that the student leading the audience through. Slides contain required volume of information. Everything on the slide relevant to what student is explaining.
Verbal explanation (20)	student has verbally explained and presented the subject matter discussed in the presentation poor manner by looking at the slides either projected or printed. Student's knowledge fluency of topic and business analytics subject domain is poor. student does not fully understand the study he/she is presenting. student states the information he/she is presenting not satisfactory way. Student is not speaking clearly, not loud enough,	student has verbally explained and presented the subject matter discussed in the presentation satisfactory manner without reading by looking at the slides either projected or printed. Student's knowledge fluency of topic and business analytics subject domain is satisfactory student understand the study he/she is presenting is satisfactory. student states the information he/she is presenting satisfactory way. Student is very well	student has verbally explained and presented the subject matter discussed in the presentation good manner without reading by looking at the slides either projected or printed. Student's knowledge fluency of topic and business analytics subject domain is good student understand the study he/she is presenting good way. student states the information he/she is presenting good way. Student is very well speaking	student has verbally explained and presented the subject matter discussed in the presentation very good.manner without reading by looking at the slides either projected or printed. Student's knowledge fluency of topic and business analytics subject domain is very good student understand the study he/she is presenting better way. student states the information he/she is presenting excellent way. Student is very well	student has verbally explained and presented the subject matter discussed in the presentation excellent manner without reading by looking at the slides either projected or printed. Student's knowledge fluency of topic and business analytics subject domain is excellent. student understand the study he/she is presenting very well. student states the information he/she is presenting excellent way. Student is very well speaking clearly, loud enough,

	and not making contact with the audience. The student is not mumbling, quiet, speaking into the screen, and hard to understand.	speaking clearly, loud enough, and making contact with the audience. The student is not mumbling, quiet, speaking into the screen, and hard to understand.	clearly, loud enough, and making contact with the audience. The student is not mumbling, quiet, speaking into the screen, and hard to understand.	speaking clearly, loud enough, and making contact with the audience. The student is not mumbling, quiet, speaking into the screen, and hard to understand.	and making contact with the audience. The student is not mumbling, quiet, speaking into the screen, and hard to understand.
Research literature review (15)	student has not done satisfactory research literature review relevant to the discussion. student has not properly recognized various statistical and geospatial software applications available as open source or licensed in the market to support the discussion. The knowledge about latest statistical and GIS software applications also exhibited not satisfactory level	student has done is satisfactory research literature review relevant to the discussion. student has satisfactory level recognized various statistical and geospatial software applications available as open source or licensed in the market to support the discussion. The knowledge about latest statistical and GIS software applications also exhibited satisfactory level	student has done good research literature review relevant to the discussion. How clearly student has recognized various statistical and geospatial software applications available as open source or licensed in the market to support the discussion. The knowledge about latest statistical and GIS software applications also exhibited better way.	student has done very good research literature review relevant to the discussion. How clearly student has recognized various statistical and geospatial software applications available as open source or licensed in the market to support the discussion. The knowledge about latest statistical and GIS software applications also exhibited very good way.	student has done excellent research literature review relevant to the discussion. How clearly student has recognized various statistical and geospatial software applications available as open source or licensed in the market to support the discussion. The knowledge about latest statistical and GIS software applications also exhibited very well.
Importance of the topic discussion (25)	The student has not discussed satisfactory level the importance of the topic as per global/local trends. Identification of the issues/opportunities and solutions related to the selected topic with the support of statistical, geospatial tools, technologies and methodologies. Feasibility of the proposal also has not emphasized.	The student has discussed satisfactory level the importance of the topic as per global/local trends. Identification of the issues/opportunities and solutions related to the selected topic with the support of statistical, geospatial tools, technologies and methodologies. Feasibility of the proposal also has emphasized satisfactory level.	The student has discussed better way the importance of the topic as per global/local trends. Identification of the issues/opportunities and solutions related to the selected topic with the support of statistical, geospatial tools, technologies and methodologies. Feasibility of the proposal also has emphasized better way.	The student has discussed better way the importance of the topic as per global/local trends. Identification of the issues/opportunities and solutions related to the selected topic with the support of statistical, geospatial tools, technologies and methodologies. Feasibility of the proposal also has emphasized very good way.	The student has discussed excellent way the importance of the topic as per global/local trends. Identification of the issues/opportunities and solutions related to the selected topic with the support of statistical, geospatial tools, technologies and methodologies. Feasibility of the proposal also has emphasized excellent way.

Further Information

Who can answer questions about my assessment?

Questions about the assessment should be directed to the staff member who has set the task/assessment brief. This will usually be the Module Leader. They will be happy to answer any queries you have.

Staff members can often provide feedback on an assignment plan but cannot review any drafts of your work prior to submission. The only exception to this rule is for Dissertation Supervisors to provide feedback on a draft of your dissertation.

Referencing and independent learning

Please ensure you reference a range of credible sources, with due attention to the academic literature in the area. The time spent on research and reading from good quality sources will be reflected in the quality of your submitted work.

Remember that what you get out of university depends on what you put in. Your teaching sessions typically represent between 10% and 30% of the time you are expected to study for your degree. A 20-credit module represents 200 hours of study time. The rest of your time should be taken up by self-directed study.

Unless stated otherwise you must use the **HARVARD** referencing system. Further guidance on referencing can be found in the Study Smart area on Moodle and at www.citethemrightonline.com (use your university login details to access the site). Correct referencing is an easy way to improve your marks and essential in achieving higher grades on most assessments.

Technical submission problems

It is strongly advised that you submit your work at least 24 hours before the deadline to allow time to resolve any last minute problems you might have. If you are having issues with IT or Turnitin you should contact the IT Helpdesk on (+44) 2920 417000. You may require evidence of the Helpdesk call if you are trying to demonstrate that a fault with Moodle or Turnitin was the cause of a late submission.

Extensions and mitigating circumstances

Short extensions on assessment deadlines can be requested in specific circumstances. If you are encountering particular hardship which has been affecting your studies, then you may be able to apply for mitigating circumstances. This can give the teachers on your programme more scope to adapt the assessment requirements to support your needs. Extensions and mitigating circumstances policies and procedures are regularly updated. You should refer to your degree programme or school Moodle pages for information on extensions and mitigating circumstances.

Unfair academic practice

Cardiff Met takes issues of unfair practice **extremely seriously**. The University has procedures and penalties for dealing with unfair academic practice. These are explained in full in the University's Unfair Practice regulations and procedures under Volume 1, Section 8 of the Academic Handbook. The Module Leader reserves the right to interview students regarding any aspect of their work submitted for assessment.

Types of Unfair Practice, include:

Plagiarism, which can be defined as using without acknowledgement another person's words or ideas and submitting them for assessment as though it were one's own work, for instance by copying, translating from one language to another or unacknowledged paraphrasing. Further examples include:

- Use of any quotation(s) from the published or unpublished work of other persons, whether published in textbooks, articles, the Web, or in any other format, where quotations have not been clearly identified as such by being placed in quotation marks and acknowledged.
- Use of another person's words or ideas that have been slightly changed or paraphrased to make it look different from the original.
- Summarising another person's ideas, judgments, diagrams, figures, or computer programmes without reference to that person in the text and the source in a bibliography/reference list.
- Use of assessment writing services, essay banks and/or any other similar agencies (NB. Students are commonly being blackmailed after using essay mills).
- Use of unacknowledged material downloaded from the Internet.
- Re-use of one's own material except as authorised by your degree programme.

Collusion, which can be defined as when work that that has been undertaken with

others is submitted and passed off as solely the work of one person. Modules will clearly identify where joint preparation and joint submission are permitted, in all other cases they are not.

Fabrication of data, making false claims to have carried out experiments, observations, interviews or other forms of data collection and analysis, or acting dishonestly in any other way.

How is my work graded?

Assessment grading is subject to thorough quality control processes. You can view a summary of these processes on the [Assessment Explained Infographic](#).

Grading of work at each level of Cardiff Met degree courses is benchmarked against a set of general requirements set out in [Volume 1, Section 4.3](#) of our Academic Handbook. A simplified version of these Grade Band Descriptors (GBDs) with short videos explaining some of the academic terminology used can be accessed via the [Facilitation of Learning](#) resource page.

We would strongly recommend looking at the [Study Smart](#) area of Moodle to find out more about assessments and key academic skills which can have a significant impact on your grades. Always check your work thoroughly before submission.

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